

- **7. DESIGN CATEGORIES**

- The project uses five basic categories.

- **7.1 Floral**

- Floral sarees contain flower, leaf, or flower-like patterns.

- **7.2 Geometric**

- Geometric sarees contain patterns based on shapes such as circles, squares, triangles, diamonds, and other repeated geometric structures.

- **7.3 Paisley**

- Paisley designs contain traditional curved or teardrop-shaped motifs.

- **7.4 Plain**

- Plain sarees contain little or no major repeated decorative pattern.

- **7.5 Striped**

- Striped sarees contain repeated horizontal, vertical, or diagonal lines.

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- **8. DATASET**

- The dataset contains saree images organized according to their design.

- The dataset can be organized as:

- dataset/

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- train/

- floral/

- geometric/

- paisley/

- plain/

- striped/

-

- validation/

- floral/

- geometric/

- paisley/

- plain/

- striped/

-

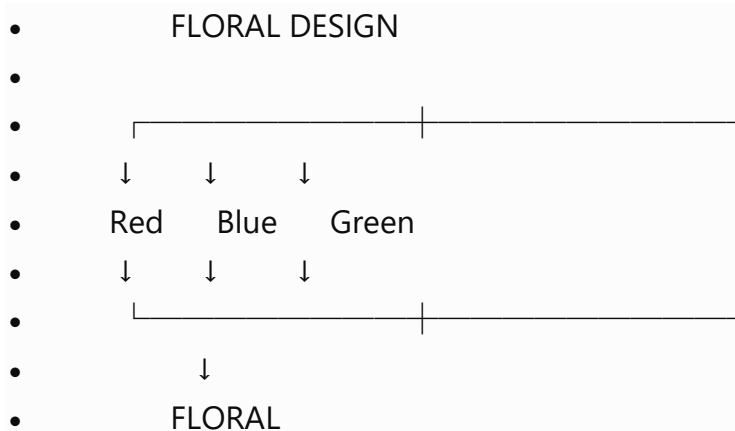
- test/

- floral/

- geometric/
- paisley/
- plain/
- striped/
- For an initial project, approximately 100 images per category can be used.
- Therefore:
- 5 categories × 100 images
- = approximately 500 images
- A larger dataset can be used to improve the model.

• 9. COLOUR-INVARIANT CONCEPT

- The main concept of the project is colour invariance.
- Colour invariance means that the model should identify the design even if the colour changes.
- For example:



- The model should focus on:
- Shapes
- Texture
- Motifs
- Pattern arrangement
- Repeated structures
- rather than depending only on colour.

• 10. DATA AUGMENTATION

- Data augmentation creates slightly different versions of training images.
- The project can use:

- Horizontal flipping
- Rotation
- Zoom
- Contrast changes
- Brightness changes
- Saturation changes
- Colour variations
- For example:
- Original Image
- ↓
- Brightness Change
- ↓
- Contrast Change
- ↓
- Saturation Change
- ↓
- Rotation
- ↓
- Training Images
- This gives the model more variation during training.

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- **11. DEEP LEARNING MODEL**
 - The project uses **MobileNetV2**.
 - MobileNetV2 is a lightweight convolutional neural network designed for image-related tasks.
 - Instead of training a complete neural network from the beginning, the project uses **transfer learning**.
 - The concept is:
 - Pretrained MobileNetV2
 - ↓
 - Already learned general image features
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 - Saree Dataset
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 - Adapt to saree designs

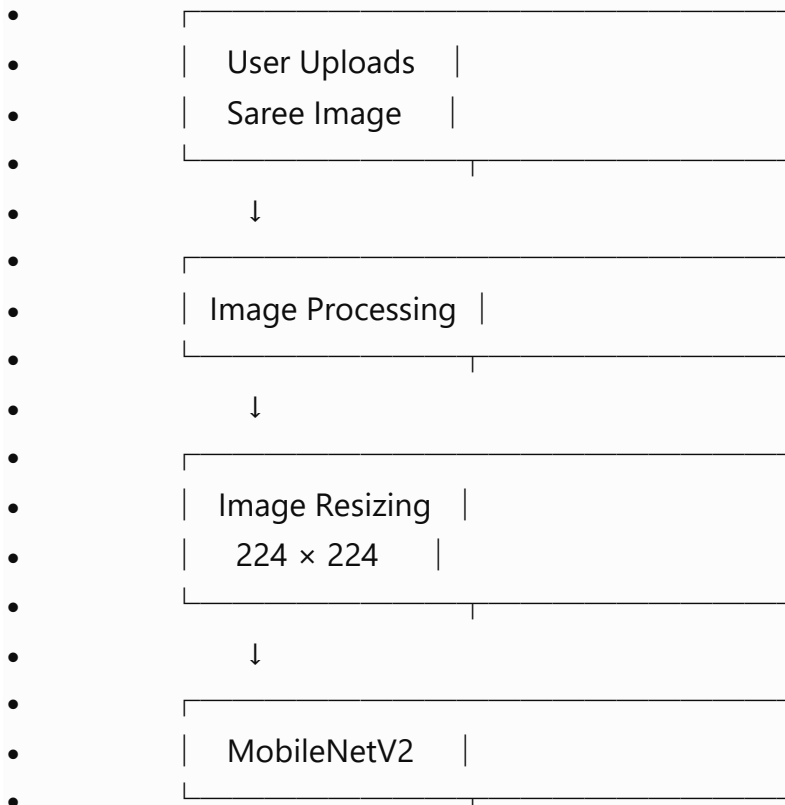
- ↓
- Saree Design Prediction
- This makes the project easier and faster to develop.

• **12. TRANSFER LEARNING**

- Transfer learning means using knowledge learned from an existing model for a new task.
- The pretrained model has already learned general visual features such as:
 - Edges
 - Shapes
 - Textures
 - Patterns
- We use these features to recognize saree designs.
- This is more practical than training a large deep-learning model completely from scratch.

• **13. SYSTEM ARCHITECTURE**

- The system architecture is:



• Software	• Purpose
• Python	• Programming
• TensorFlow	• Deep Learning
• NumPy	• Numerical operations
• Pillow	• Image processing
• Streamlit	• Web application
• MobileNetV2	• Image classification

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- **16. HARDWARE REQUIREMENTS**
 - Basic hardware can be used.
 - **Minimum**
 - Computer/Laptop
 - 4 GB RAM or more
 - Internet connection for installing libraries
 - Storage for dataset
 - **Recommended**
 - 8 GB RAM or more
 - Modern processor
 - GPU is optional for this small project

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- **17. APPLICATION WORKFLOW**
 - The application works as follows:
 - **Step 1**
 - The user opens the application.
 - **Step 2**
 - The user uploads a saree image.
 - **Step 3**

- The image is resized to:
- 224 × 224 pixels
- **Step 4**
- The image is converted into numerical data.
- **Step 5**
- The trained MobileNetV2 model processes the image.
- **Step 6**
- The model produces probabilities for each design.
- Example:
- Floral 92%
- Geometric 2%
- Paisley 3%
- Plain 1%
- Striped 2%
- **Step 7**
- The category with the highest probability is selected.
- **Step 8**
- The result is displayed to the user.

- **18. SAMPLE RESULT**

- Example input:
- Blue saree with floral pattern
- Output:
- Predicted Design: Floral
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- Confidence: 92.45%
- Another example:
- Green saree with paisley pattern
- Output:
- Predicted Design: Paisley
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- Confidence: 89.20%

- **19. COLOUR-INVARIANCE TEST**

- The main experiment can be performed using different colours of the same design.

• Saree	• Colour	• Actual Design	• Predicted
• Saree 1	• Red	• Floral	• Floral
• Saree 2	• Blue	• Floral	• Floral
• Saree 3	• Green	• Floral	• Floral
• Saree 4	• Red	• Paisley	• Paisley
• Saree 5	• Blue	• Paisley	• Paisley
• Saree 6	• Green	• Paisley	• Paisley

- The purpose of this test is to observe whether the model continues to identify the design when colour changes.

• **20. ADVANTAGES**

- The proposed system has several advantages:
- Simple to use
- Easy image upload
- Fast prediction
- Uses transfer learning
- Requires less training compared with training from scratch
- Can recognize multiple saree designs
- Attempts to reduce colour dependence
- Can be extended with additional categories
- Can be deployed as a web application

• **21. LIMITATIONS**

- The project also has some limitations:
- Prediction depends on the quality of the training dataset.
- A small dataset may reduce accuracy.

- Very complicated patterns may be difficult to classify.
- Poor lighting can affect recognition.
- Background objects can affect the prediction.
- Similar designs may sometimes be confused.
- Colour invariance is not guaranteed for every possible colour or image condition.

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• **22. FUTURE ENHANCEMENTS**

- The project can be improved in several ways.

• **More designs**

- Add:
- Zari
- Temple
- Peacock
- Checks
- Traditional motifs
- Abstract designs

• **Saree type recognition**

- The system can also recognize:
- Kanjeevaram
- Banarasi
- Chanderi
- Mysore Silk
- etc.

• **Colour recognition**

- A separate module can identify:
- Red
- Blue
- Green
- Pink
- Yellow
- etc.

• **Mobile application**

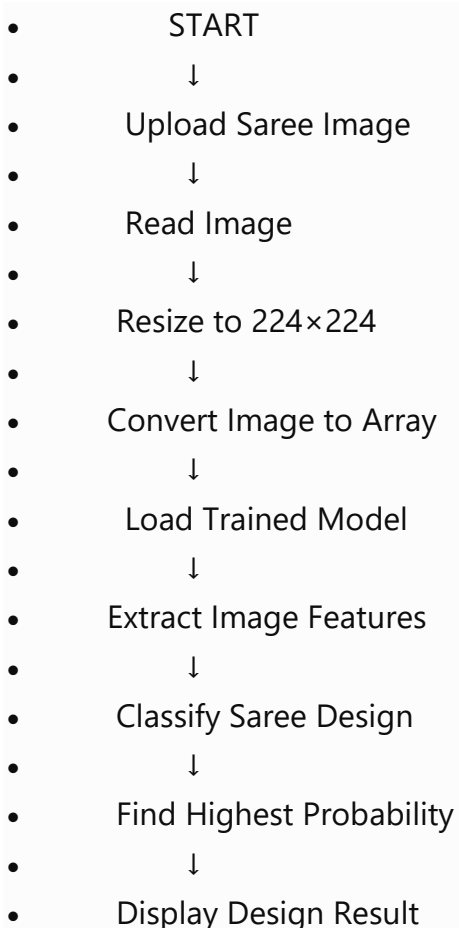
- The system can be converted into an Android or iOS application.

• **Larger dataset**

- Using thousands of saree images can improve generalization.

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- - **23. CONCLUSION**
 - The project **Color-Invariant Saree Design Recognition Using Deep Learning** demonstrates a simple approach to recognizing saree designs from images.
 - The system uses a pretrained MobileNetV2 model and image augmentation techniques to recognize five basic saree design categories: Floral, Geometric, Paisley, Plain, and Striped.
 - The main focus of the project is to reduce the influence of colour on design recognition. By providing images with different colour and visual variations during training, the model is encouraged to learn useful design-related features.
 - The developed Streamlit application provides a simple interface where users can upload a saree image and receive the predicted design and confidence score.
 - The project can be extended in the future with more saree categories, larger datasets, colour recognition, saree type recognition, and mobile deployment.

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- **24. PROJECT FLOWCHART**



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- Display Confidence Score
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- END
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